# GenAI for Audio



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# Learning Objectives for Today

- Get a high-level overview of generative AI capabilities in audio, including an introduction to text-to-speech and speech synthesis.
- Understand the basics of text analysis and how it applies to audio applications.
- Learn about linguistic processing and waveform generation techniques with practical demonstrations.
- Explore how WaveNet generates human-like speech through a live demo.
- Understand the role of GANs in creating new music and sounds.
- Gain hands-on experience with voice recognition technologies and integrating voice with LLMs through coding exercises.
- (Optional) Dive into advanced voice models and architectures for further learning.



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# Today's Agenda

| | 1st Hour (9-10 PT) | 2nd Hour (10-11 PT) | 3rd Hour (11-12 PT) | 4th Hour (12-1 PT) |
|---|---|---|---|---|
| 15-20 minutes | Brief Introduction to Text and Speech | WaveNet with demo | Voice Recognition | • (Optional) Advanced: Voice Models and Architecture |
| 30-45 minutes | • Text Analysis | | | |
| 60-90 minutes | • Linguistic Processing and Waveform Techniques with demo | GANS | Voice with LLMs with coding exercise for class | |
| 5 minutes | Summary | Break | Break | Summary |

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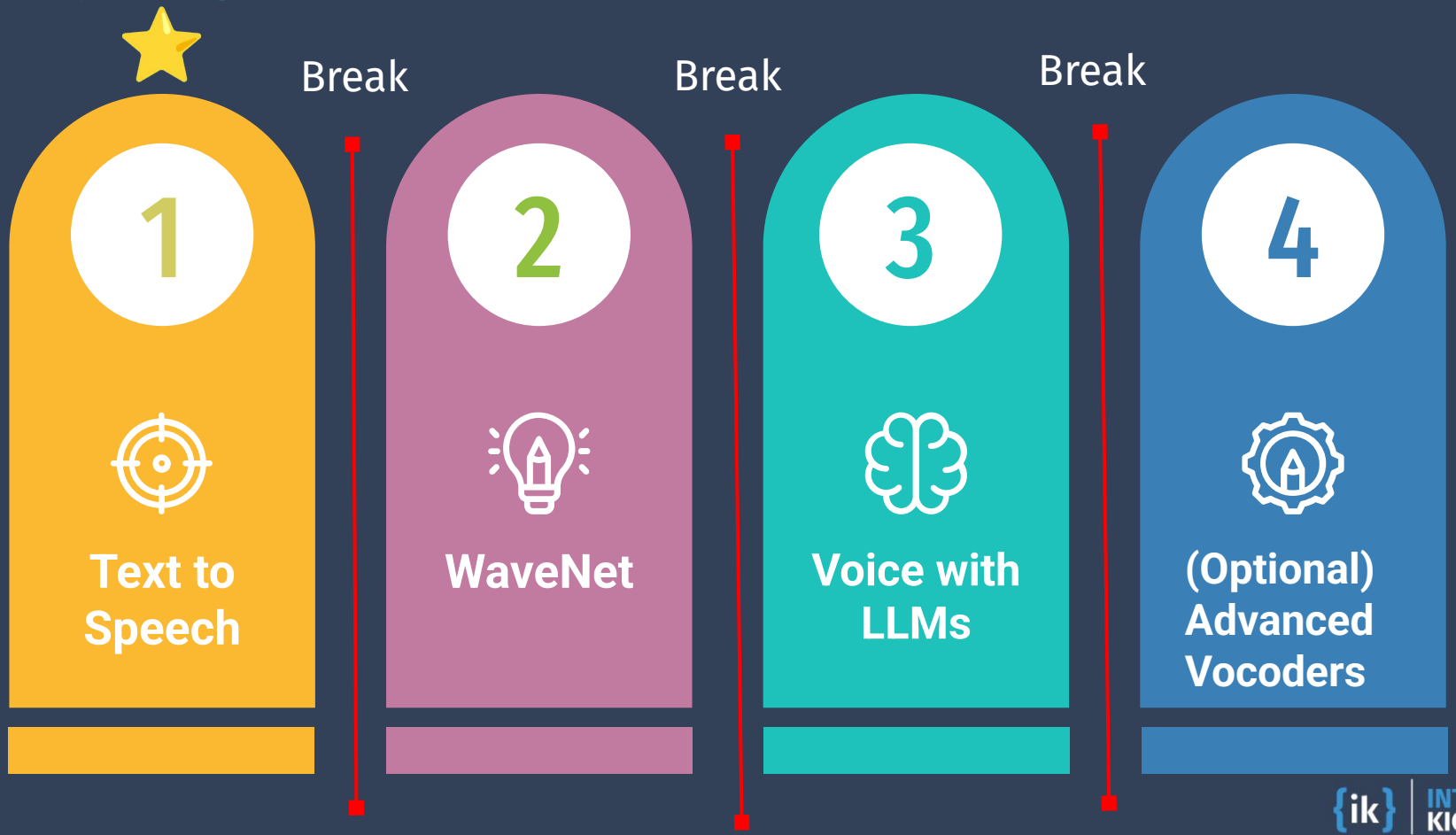
# How is Today's Content Relevant to my Role?

- **PMs:** Identify new product opportunities in voice generation and music creation with generative AI.
- **TPMs:** Plan and manage projects involving text-to-speech synthesis and deep learning models like WaveNet.
- **SDEs:** Master voice generation technologies, customize and integrate them, addressing technical and ethical considerations.
- **Engineering Managers:** Understand voice models like vocoders and Tacotron, integrate them with LLMs, and guide implementation.
- **DevOps Engineers:** Learn software and infrastructure requirements for generative AI in audio, including waveform generation, WaveNet, GANs, and LLM integration.
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# Today's Agenda




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# TOPIC #1
# Brief Introduction to Text and Speech



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# Let’s start with this example!

- An AI bot reading out loud text from a computer screen



https://www.naturalreaders.com/online/



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# Text to Speech Process

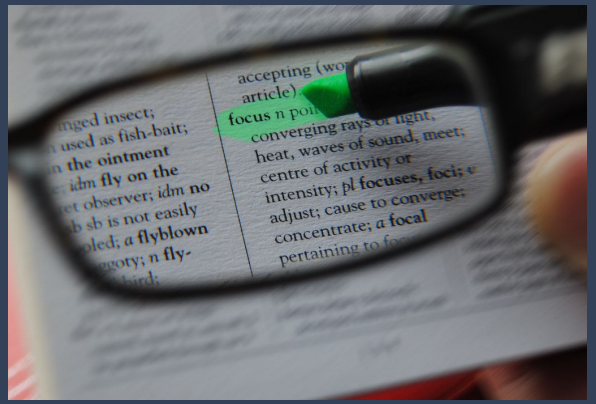
- Text Analysis
- Linguistic Processing
- Waveform Generation

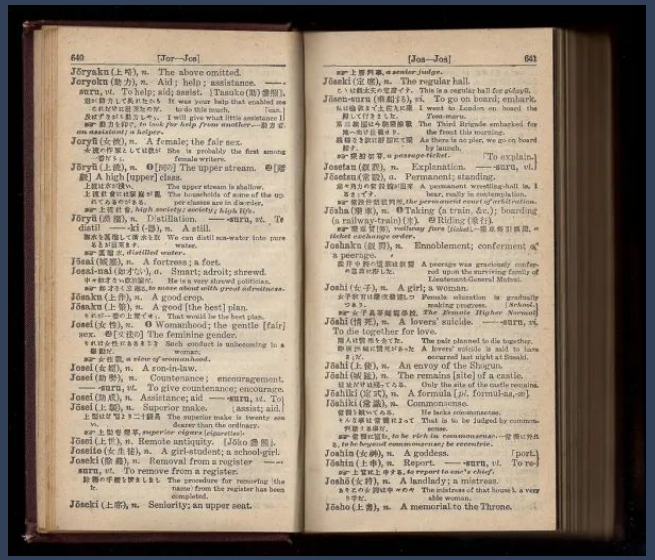


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# Text Analysis

How does the following sentence get broken down?
Sentence: Dr. Swamy prescribed 30 mg of medicine.



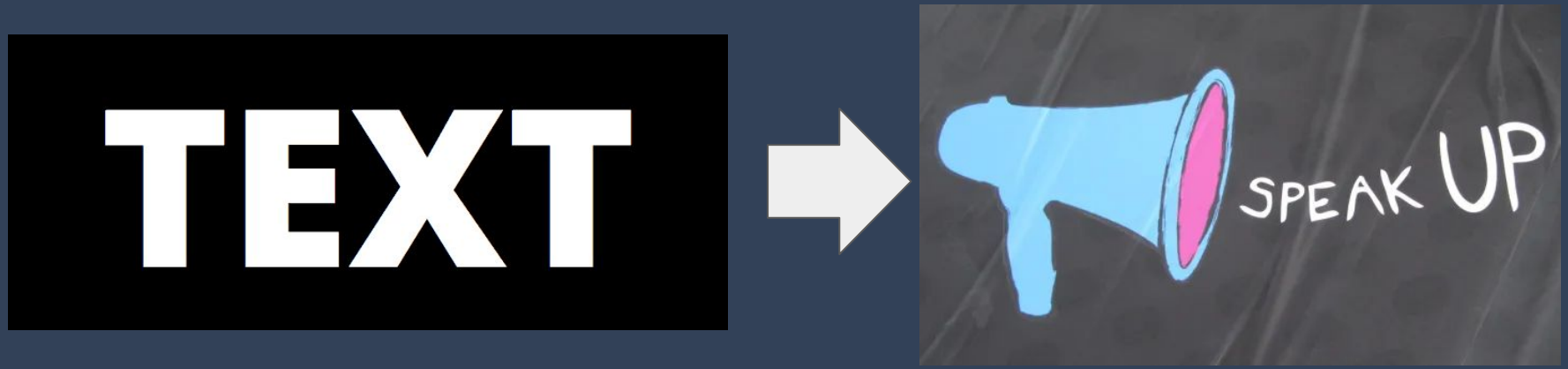






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# Linguistic Processing

- Word gets converted to phonetic pronunciation.






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# How it works

- The sentence is: Read the book that student read last night. How does AI know how to say this? Good training on both mispronounced and correctly pronounced words in different contexts









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# Limitations

- AI doesn’t fully understand emotions or sarcasm.
- AI doesn’t always have the latest slang









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# How to overcome these limitations?

More training







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# Summary

01 Part of Speech(POS) Tagging of text

02 Understanding words in the context

03 Converting words to phonetic pronunciations

04 Waveform generation for phonetic pronunciations

05 Generating Speech



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QUIZ
TIME!

# Recap Quiz

**Which of the following is NOT a use of Text-to-Speech technology?**

**A** Reading out loud the content of a webpage

**B** Converting spoken words into written text

**C** Announcing navigation directions in cars

**D** Helping with pronunciation in language learning apps



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# Recap Quiz - Solution

**Which of the following is NOT a use of Text-to-Speech technology?**

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- **B** Converting spoken words into written text
- **C** Announcing navigation directions in cars
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# Recap Quiz

**For whom is Text-to-Speech technology especially beneficial?**

**A** People with visual impairments

**B** Individuals who have difficulty reading

**C** Language learners

**D** All of the Above



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# Recap Quiz - Solution

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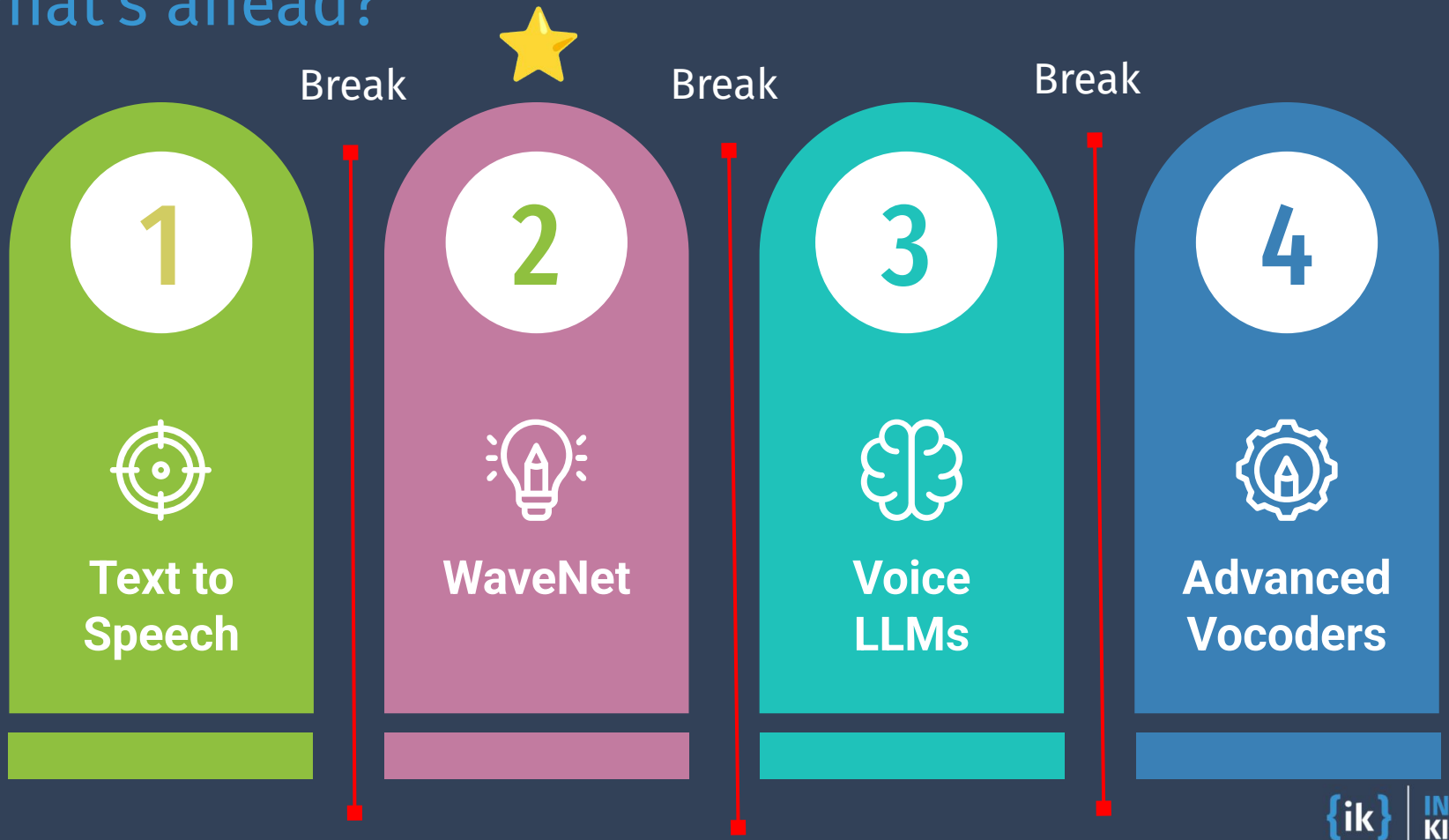
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# What's ahead?






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# Speaker Diarization



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# What is Speaker Diarization?

- Speaker diarization is the process of partitioning an audio stream into segments to identify the identity of the speaker.
- How? Identify and label segments of audio that correspond to different speakers which helps identify different speakers in a conversation



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# Key Steps in Speaker Diarization

Voice Activity Detection: identifies segments of audio that contains speech and distinguishes that from non-speech segments such as background noise
Feature Extraction: identifies features that can represent different speakers
Segmentation: divides audio into segments that are likely to contain speech from a single speaker
Clustering: grouping segments of audio that belong to the same speaker
Re-segmentation: Refine initial segmentation and clustering to improve accuracy



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# Case Study: Call Center

- Problem: Call centers often struggle with high volumes and need to provide consistent, accurate responses to customer inquiries.
- Solution: Use conversational AI with audio to improve efficiency and customer satisfaction.



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# How would this be implemented?

Input Audio: a recorded conversation involving multiple speakers
Voice Activity Detection: identify segments containing speech or silence
Feature Extraction: Extract relevant features of each speech segment
Create segmentation: divide speech into smaller segments
Clustering: group audio segments based upon similarities
Re-segmentation and refinement: refine segments and improve clustering
Output: labeled segments indicate which parts of audio belong to which speaker(call center agent or customer)



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# Benefits of this

- Improved call transcription and analysis
- Enhanced customer experience
- Better understanding of customer sentiment
- More personalized customer service



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# TOPIC #2
# WaveNet



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# Let’s start with this example!

- WaveNet generating speech
- https://cloud.google.com/text-to-speech/docs/voices



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# What does WaveNet do?

Google WaveNet creates speeches or can translate text.
It can also create music.







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# What is this all about?

WaveNet is an autoregressive model. It looks at a sequence of sounds and predicts what the next sound will be. How?
WaveNet trains from a training set of various different types of sounds to product to infer what type of sound it needs to produce.









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# Limitations

While WaveNet can produce nice speeches and music, it can also be misused if we are not careful. If trained on a small training sample, it can misrepresent how certain people can sound. It can also portray an accent or dialect of a language incorrectly if not trained on enough people.



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# How to overcome these limitations?

Ensure WaveNet is trained on on adequate training dataset of various sounds to ensure the voices it speaks with are correctly represented.






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# Summary

| | |
|---|---|
| 01 | WaveNet generating speeches and music |
| 02 | Receives input of sounds and predicts the next sound |
| 03 | Can potentially misrepresent how people sound |
| 04 | Can potentially make music that isn't appealing |
| 05 | Train WaveNet Model on an ample datasets of sound and music |



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QUIZ
TIME!

# Recap Quiz

**What is a unique capability of WaveNet technology in the field of music?**

**A** It can automatically tune musical instruments

**B** It synthesizes realistic human singing voices from text

**C** It increases the volume of live performances without additional speakers

**D** It identifies the genre of music based on rhythm patterns



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# Recap Quiz - Solution

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# Recap Quiz

**Who developed WaveNet technology?**

- **A** Microsoft
- **B** Google’s DeepMind
- **C** Apple
- **D** Amazon



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# Recap Quiz - Solution

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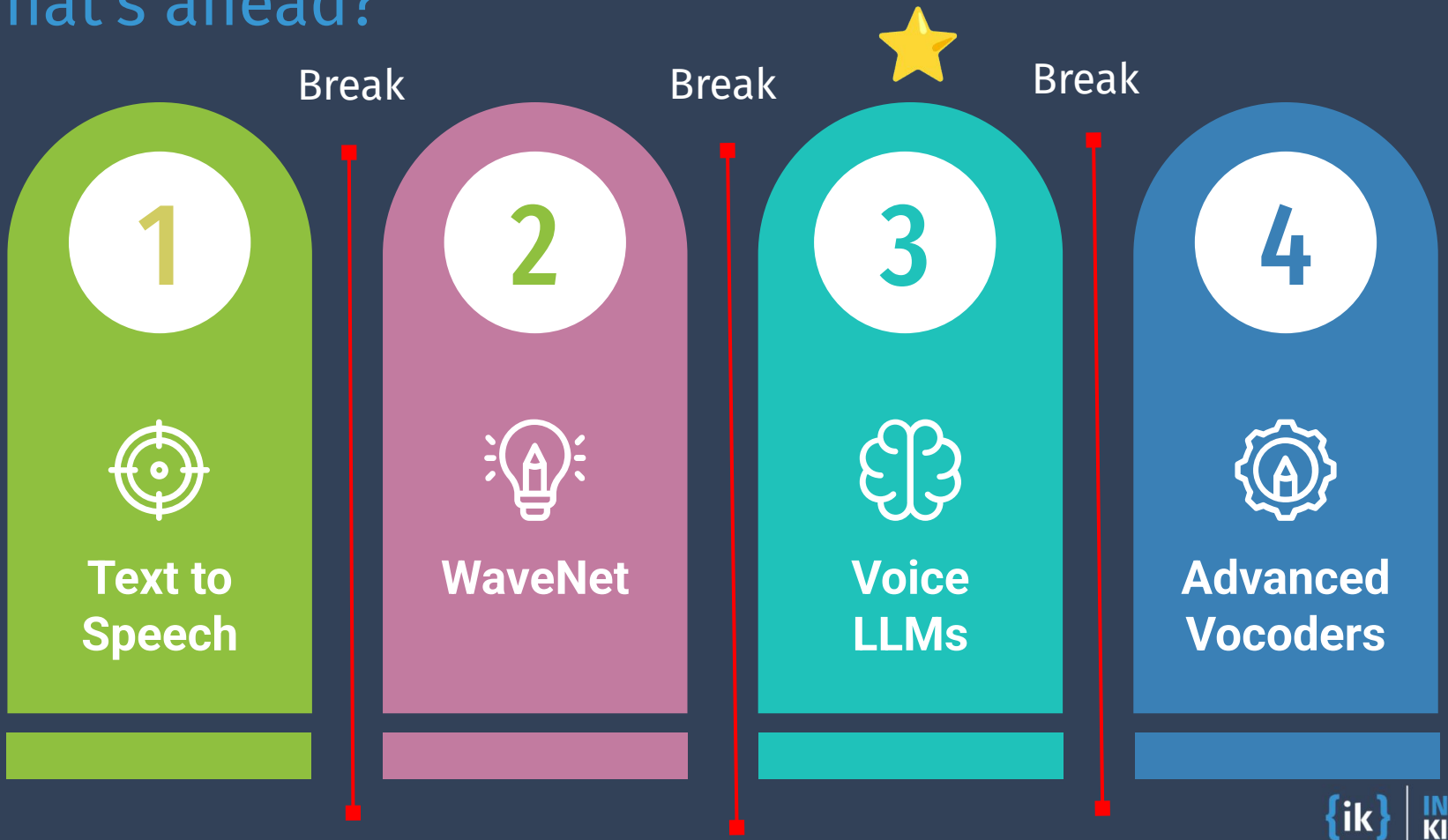
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# What's ahead?






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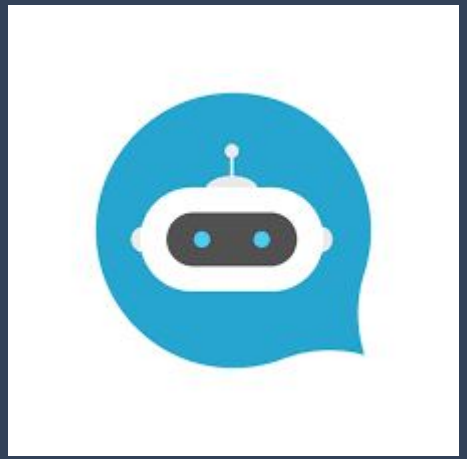
# TOPIC #3
# Voice with LLMs



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# LLMs with Voice

LLMs with voice are a way for people to communicate effectively with a computer.
Having such capabilities can help improve the user experience for people to do their various daily tasks.





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# Applications of LLMs with Voice

Alexa
Siri
Google Voice Assistant








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# Prior to genAI for Audio

Before GenAI, Siri had concatenative synthesis and parametric synthesis. Both had many limitations



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# Concatenative Synthesis

Early on Siri and Google Voice would concatenate pre-recorded audio segments from a large database of speech fragments.
While this produced great audio, it didn't sound natural and lacked flexibility.



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# Parametric Synthesis

Later, Siri and Google Voice utilized parametric synthesis models which were statistical models to generate speech.
These systems would generate audio waveforms by adjusting parameters of a pre-defined model based on input text
While this was an improvement over concatenative synthesis when it comes to flexibility, the voices would still sound robotic and less natural.



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# How were LLMs with voice an improvement?

- Deep Learning Models that were autoregressive models that predicted audio samples based upon previous audio samples resulting in more natural sounding audio
- Transformers helped handle sequential data better by using self-attention mechanisms to produce more natural and coherent speech
- Transformers captured the bigger picture of long-term relationships within a piece of audio better than traditional models such as LSTM.
- Transformers were much faster at processing information quickly than older models.



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# Limitations

- Can not understand different accents or dialects or different types of voices
- Do not work well if a person asks the chat bot in a busy street with lots of noise
- Sometimes don’t understand the context of the user’s question





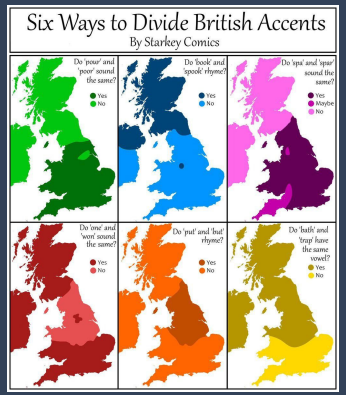

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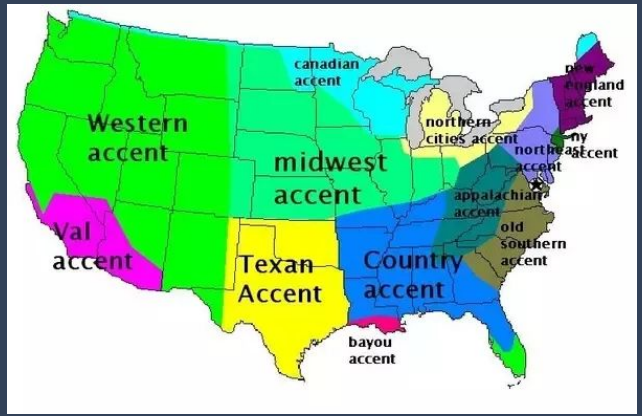
# How to overcome these limitations?

- Wider range of accents, dialects, and languages are used in the training datasets for LLMs with voice
- Noise cancellation and background noise suppression technologies
- Using GPUs for Machine Learning to improve the time it takes the LLM to run and process what is said

Vocabulary

| Canada | American | British |
|---|---|---|
| bus depot | bus station | coach station |
| Elevator | Elevator | Lift |
| Gas | Gas | Petrol |
| main floor | first floor | ground floor |
| phone, call (v) | call | Phone |
| Vacation | Vacation | holiday |
| Washroom | Ladies' room | Gents/Ladies |
| University | College | University |
| Railways | Railroads | Railways |
| Fire hall | Fire house | Fire station |






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# Fake Audio Detector

- A fake audio detector analyzes audio data to identify and classify audio content as legitimate or fake.
- What is fake audio?
- Fake audio is misinformation, disinformation, and mal-information presented in audio formats that can mislead listeners.
- The model uses its training to assess the authenticity and reliability of the audio content.



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# Problem Statement

- Use case: you are part of a team that has noticed an uptick of fake audio on your streaming videos site.
- Product folks and TPM: envision and integrate features such as real-time fake audio detection, personalized audio experiences, and enhanced voice interfaces
- Devops engineers: enables them to design and deploy systems for the computational load of processing and generating audio and work with ML teams to do these deployments



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# Demo

Here is the link to the live coding demo:
code



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# Summary

01 LLMs allow you to talk to a computer or electronic device.

02 Other benefits of LLMs with voice

03 Examples of LLMs with voice: Alexa and Siri

04 Difficulties LLMs with Voice have understanding different types of voicess

05 Train LLMs to better understand a wider variety of voices



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QUIZ
TIME!

# Recap Quiz

**Why might LLMs with voice be useful for in smart home devices?**

**A** They can physically clean and organize a home

**B** They allow users to control devices through voice commands

**C** They improve the Wifi signal throughout the home

**D** They reduce the cost of home insurance



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# Recap Quiz - Solution

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# Recap Quiz

**What is a key feature of voice-enabled Large Language Models?**

**A** They can sing songs

**B** They can convert text into spoken words

**C** They can understand and respond to human speech

**D** Both B and C are correct



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# Recap Quiz - Solution

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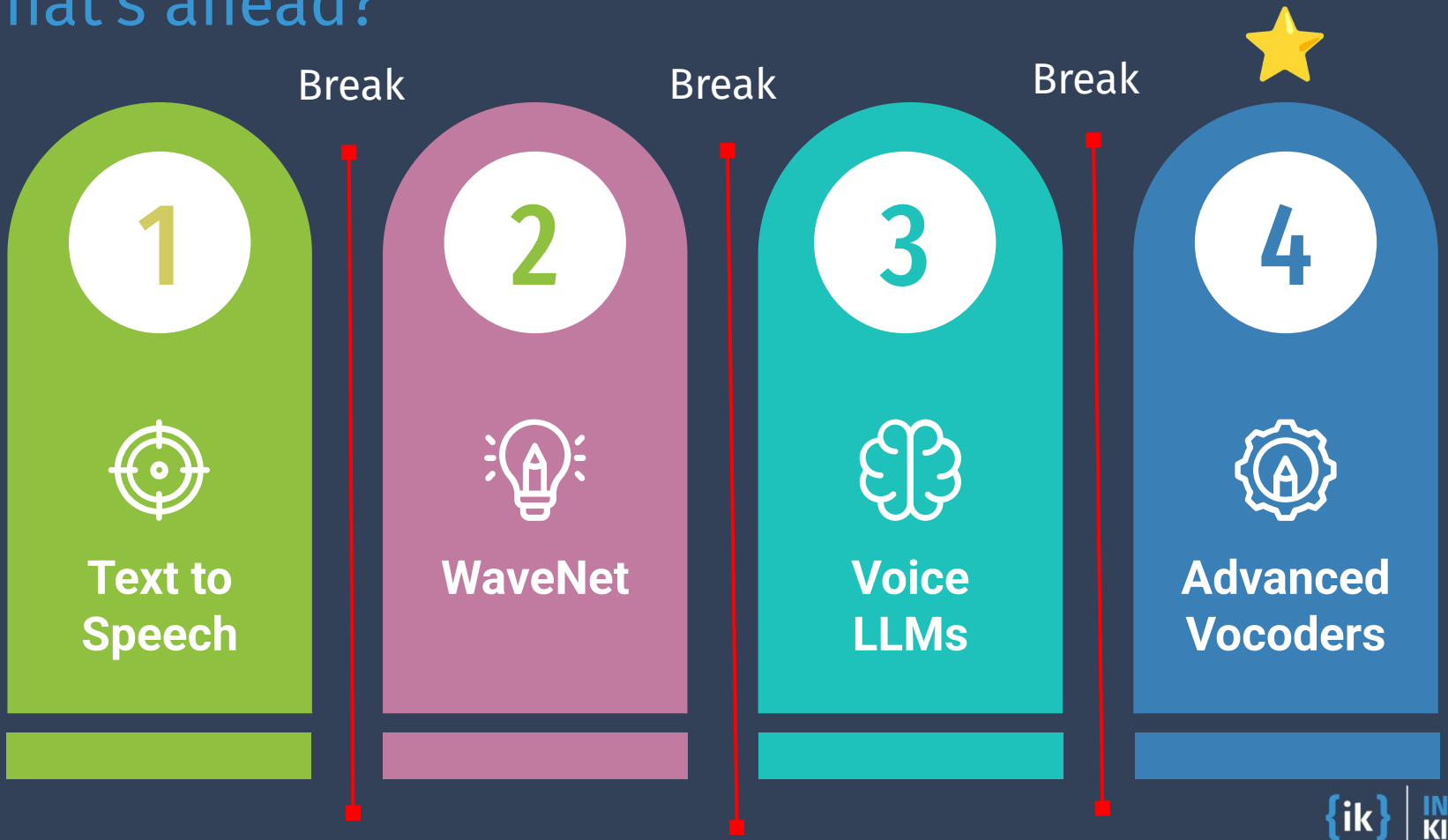
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# What's ahead?






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# TOPIC #4
# Vocoders



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# Change your voice

- Use this application to change your voice: https://voicechanger.io/



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# How vocoders work?

Vocoders analyze the human voice and split the voice into different frequencies and then put them together.





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# Applications

Music production
Acting
Radio broadcasting
Speech impairment assistance











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# Limitations

Voices can sound artificial.
Voices can not capture the true emotion of a speaker well.





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# How to overcome these limitations?

Better training of different voices
Better signal processing to filter out noise and other extraneous background fluff








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# Summary

| | |
|---|---|
| 01 | Vocoders changing your voice |
| 02 | How vocoders work |
| 03 | Real world applications of vocoders |
| 04 | Limitations of vocoders: voice can sound too artificial |
| 05 | Improvements: better signal processing and better trained samples |



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QUIZ
TIME!

# Recap Quiz

**What is a common use case of vocoders in the entertainment industry?**

**A** Creating visual effects in movies

**B** Designing costumes for theater productions

**C** Generating unique voice effects in music or movies

**D** Writing scripts for television shows



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# Recap Quiz - Solution

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# Recap Quiz

**Vocoders are commonly used in which of the following technologies?**

**A** Music Production

**B** GPS devices

**C** Digital Cameras

**D** Both A and B



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# Recap Quiz - Solution

**Vocoders are commonly used in which of the following technologies?**

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# Class Summary

| | | | |
|---|---|---|---|
| **Text to Speech** | Text Analysis | Converting text to phonetic pronunciations | Waveform generation of speech |
| **WaveNet** | speeches and music examples | How WaveNet works | WaveNet limitations |
| **Voice with LLMs** | Alexa and Siri | How voice with LLMs work | Limitations of Voice with LLMs |
| **Vocoders** | Changing your voice | How vocoders work | Limitations of vocoders |



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